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Deuterium between 31 and 1021 ppm in dietary water has no effect on lifespan in the adult housefly (*Musca domestica*)

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ABSTRACT

Deuterium, an isotope of hydrogen, exists naturally in the environment and body water at around 89–158 ppm. Reports of the health impacts of the level of deuterium in ingested water are often contradictory. We investigated the lifespan effects in houseflies fed diets made up with a range of deuterium, depleted or enriched relative to natural abundance water (31 ppm to 1021 ppm). Control houseflies were fed diets prepared with Aberdeen (57 ^{o}N) tap water (147 ppm). Pupae were planted individually to vials containing appropriately prepared deuterium diets. Day of eclosion, death and body weight at death were recorded. Deuterium levels of water, water from the diet and the uptake of dietary deuterium by the houseflies were confirmed using laser spectroscopy. Deuterium isotope enrichments of body water of the houseflies were correlated with the deuterium enrichment of the water in their food. However, the dietary effect was strongly attenuated. The lowest deuterium content in housefly body water was 67 ppm (for the 31 ppm water) and the highest 808 ppm for the 1021 ppm water. Levels of deuterium, between 31 and 1021 ppm, had no significant effect on lifespan, mortality risk or death body weight of the houseflies. Body weight at time of death did not vary with the levels of deuterium in either sex. In conclusion, using the housefly as a model, we did not find any support for the previous suggested impacts of deuterium content of the ingested water (between 31 and 1021 ppm deuterium) on survival or lifespan.

1. Introduction

Over the 90+ years since deuterium's discovery (Urey et al., 1932) there have been numerous studies of the impacts of both depleted and enriched deuterated water, on health and lifespan in various species. Before detailing the outcomes, we first discuss the action of deuterium. Hydrogen naturally occurs as 3 isotopes: protium with just one proton in the nucleus, deuterium with a proton and a neutron which is stable and the rare and radioactive tritium with a proton and two neutrons. Deuterium is constant at around 155.76 ppm in the deep ocean (Vienna Standard Mean Ocean Water (VSMOW) reference (Hagemann et al., 1970), depleted in waters more distant from the ocean and in polar regions but enriched in the tropics. The abundance of deuterium ranges between 89 in the ice of Antarctica to 158.2 ppm in surface water of the Dababa region of Chad (Basov et al., 2019). A higher abundance of up to 166.4 ppm was reported in landfill leachate (Hackley et al., 1996). Because animals and plants take up local water, the enrichment of deuterium in their body water closely matches the enrichment in the local water supply.

Deuterium has an atomic mass double that of protium. Hence of all the elements it has the greatest isotopic mass difference. This influences both the physical and chemical properties of molecules containing deuterium, as opposed to protium (Koletzko et al., 1997). Heavier molecules generally require more energy to induce them to change phase from liquid to gas. This leads to D_2O having a greater boiling point at 101.4 °C. Conversely, differences in the latent heat (Koletzko et al., 1997) of fusion (to transform from solid to liquid) mean that deuterated water ice doesn't melt until it is raised to 3.82 °C. Given the

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different latent heats of vaporisation, water containing protium ($^1H^1H^{16}O$) evaporates more readily than water containing deuterium ($^2H^2H^{16}O$). This leads to an enrichment difference between liquid water and water vapour of about 7 to 10% (depending on temperature).

In addition, to physical effects there are also impacts on chemical reactions stemming from differences in the rate constants of chemical reactions due to the mass differences of the isotopes, sometimes called the kinetic isotope effect (Katz and Crespi, 1970; Li and Snyder, 2016a). In theory the impact of the deuterium to hydrogen mass difference may slow reactions by a factor of between 6 and 18 fold (Li and Snyder, 2016a; Krumbiegel, 1991). Work in the late 1960s and early 1970s for example showed profound effects of D_2O on the timing mechanism of the circadian clock in mammals (Katz et al., 1962; Thomson, 1963), and later in invertebrates (Pittendrigh et al., 1973) and plants (Brenner and Engelmann, 1973). For example, when small mammals were given 20% deuterated water (200,000 ppm deuterium) in their drinking water their circadian rhythms lengthened by 0.9 to 1.5 h (Daan and Pittendrigh, 1976).

Li and Snyder (Li and Snyder, 2016b) observed that as yeast aged, the levels of heavy isotopes of hydrogen, carbon and nitrogen in amino acids were diminished. Repleting these by supplementing the diet with heavy isotopes extended the replicative lifespan of the yeast by up to 85% (Li and Snyder, 2016b). Generally, increasing levels of deuterium leads to a reduction in rates of reaction with numerous positive downstream impacts. Several studies have documented beneficial impacts on various metabolic processes including cancer and heart disease. In the 1980s it was shown that drinking water with 30% deuterium (300, 000 ppm) slowed the growth of human cancers transplanted into nude mice to between 22 and 65% of the size in controls (Altermatt et al., 1987; Altermatt et al., 1988) and improved their survival time (Laissue et al., 1982). Providing rats fed a high salt diet with 25% deuterated water (D_2O) in their drinking water reduced hypertension, because the D_2O acted as a calcium channel blocker and normalized calcium uptake in vascular smooth muscle (Vasdev et al., 1990a; Vasdev et al., 1990b; Vasdev et al., 1991). Adding 7.5% and 15% D_2O to the diet of developing *Drosophila* significantly extended mean lifespan with no negative impacts on fecundity. However, higher doses (22.5%) had several negative impacts including prolonged development and reduced viability (Hammel et al., 2013). A separate study showed survival was negatively impacted by 50% deuterated water in *Drosophila* (Samis et al., 1974).

These beneficial effects were all manifest at very high levels of enrichment, between 75,000 ppm and 300,000 ppm. There are fewer studies of the effects of much lower levels of deuterium. In the 1930s studies exploring the effects of 600 ppm deuterium (4× the normal background) suggested various lower organisms, including flatworms and Euglena, grew faster (Barnes, 1934; Barnes and Larson, 1935). Later studies in the 1990s extended these growth enhancing effects of 600 ppm deuterated water to human fibroblasts (Somlyai et al., 1993). Zhang et al. (Zhang et al., 2020) showed that 350 ppm deuterium in water accelerated the growth of three different human cancer cell lines by reducing the production of ROS in mitochondria.

First shown more than thirty years ago, it was reported that giving water that has levels of deuterium below the level of natural abundance (deuterium depleted water or DDW) at 30–40 ppm inhibited the growth of tumours transplanted into mice and significantly increased the survival times of the host animals (Somlyai et al., 1993). This was an unexpected outcome and has led to many studies exploring the potential of DDW as an anticancer agent (Gyöngyi and Somlyai, 2000; Hartmann et al., 2005; Cong et al., 2010; Somlyai et al., 2010; Yavari and Kooshesh, 2019; Boros et al., 2021; Lu and Chen, 2024). In general, these studies have confirmed that DDW can inhibit the growth of multiple tumour cell lines. These effects have been linked to elevations in the activities of superoxide dismutase and catalase enzymes (Yavari and Kooshesh, 2019). Other studies have shown oxidative stress induced by tumour growth was reduced in the liver of mice following 15 days of DDW at 85 ppm. Specifically, giving DDW to the tumour group increased activity of catalase, decreased glutathione peroxidase and reductase activity and reversed protein carbonyl levels (Bayrak et al., 2022).

In addition to its anti-cancer effects there have also been suggestions that DDW may have neuroprotective, anti-ageing, anti-diabetic and anti-obesity effects (reviewed in (Qu et al., 2024)). There have been some limited human trials on the effects of DDW with variable outcomes. Four biological mechanisms were suggested to underlie the potential anti-cancer effects of DDW, presumably also mediating its other impacts (Qu et al., 2024). 1) the suppression of proliferation and division of tumour cells, 2) the induction of tumour cell apoptosis, cell cycle arrest, and the expression of cell cycle-related regulatory proteins, 3) inhibition of the expression of genes associated with tumour occurrence, development, and prognosis and 4) induction of an imbalance between mitochondrial ROS production and antioxidant neutralisation which leads to cellular oxidative stress and thereby results in slowed cell growth and activation of apoptosis.

Whether deuterium depletion or enrichment confers beneficial effects remains unclear. Biological effects have been reported across a wide range of deuterium concentrations, ranging from depleted levels (30 ppm) to highly enriched levels (300,000 ppm). These findings highlight the complexity of the response to deuterium and the need for further research to clarify the biological effects of deuterium across different concentrations leading to extended lifespan. Here we investigate the effects of deuterium depletion and enrichment in dietary water (between 31 and 1021 ppm) on the body weight at time of death, and lifespan of the housefly *(Musca domestica)*. Houseflies are a convenient and tractable model offering several practical advantages in lifespan studies, specifically a relatively short lifespan.

2. Methods

2.1. Stock housefly colony and rearing

A stock colony of houseflies, established from pupae (Blades Biologics Ltd), were maintained in a mesh insect rearing cage (50x50x50 cm, BugDorm-4E4545) within a climate-controlled room (25 °C, 35% humidity, 12:12 h light:dark light cycle). Adult houseflies were provided ad libitum access to a 1:5 mixture of hydrolysed yeast and sugar. Water was available via a moist sponge secured through a central slit in a plastic tub. The stock colony underwent breeding cycles of ~3 weeks. Eggs were collected by introducing a plastic tub containing a bran-based diet (70 g Organic Wheat Bran, 6 g Nestle Instant Full Cream Milk Powder, 15 g Alfalfa (blended, from Supreme Science Selective Guinea Pig Pellets mixed with 200 ml water) to the stock colony rearing cage. Females directly laid their eggs on the bran-diet and after ~16 h the tub was removed to a separate mesh cage. Larvae hatched within 24 h and developed over 7–10 days. During development the diet was maintained at ~50% moisture to prevent dehydration. Once pupae formed, the diet was allowed to dry and adults eclosed ~1 week later. Egg collection was repeated at ~5 days of age. Prior to experiment the colony were maintained for >10 generations in controlled laboratory conditions.

2.2. Experimental conditions

DDW was sourced from Preventa (25 ppm) and used to make up the depleted deuterium water by mixing appropriately with Aberdeen tap water (measured 147 ppm). Enriched treatments, were made using deuterium oxide 99.8% (Sercon). Water was prepared to the 7 x target deuterium concentrations (25, 50, 100, 150 (tap water), 250, 500, and 1000 ppm). The agar diet was composed of 8% *w/v* hydrolysed yeast, 1.5% *w/v* agar, 5% *w/v* sucrose, 0.8% *v*/v methylparaben (Nipagin). Batches of diet were made up weekly using the prepared water. Deuterium levels were verified in prepared water and diet (See Section 2.3 below).

Individual pupae were transferred to plastic vials (25 × 95 mm)

containing approximately 5 ml of agar diet. Vials were sealed with cotton wool to prevent escape. Houseflies were transferred to fresh diet every 7–10 days. Seventy-five pupae were planted per deuterium treatment (total 525) and maintained under the same conditions as stock colony. All houseflies were checked daily with day of eclosion (Day 0) and death recorded. Upon death, flies were weighed, sexed and stored in labelled Eppendorf tubes at −20 °C.

2.3. Deuterium content of prepared water, diet and body water of houseflies

The deuterium abundance of the water used for diet preparation and of water extracted from the prepared solid diets were verified, along with the isotopic enrichment of body water extracted from houseflies. For each diet treatment, ten flies that died at approximately median lifespan (46 to 56 days) were analysed. Body water was extracted by vacuum distillation (Nagy, 1983), and the resulting distillate was analysed. Isotopic analyses were performed blind using a Liquid Water Isotope Analyser (Los Gatos Research, USA) (Berman et al., 2012) equipped with a PAL LC-xt autosampler. Each sample was injected 18 times (1 μL) with only stable measurements retained for analysis. Samples were analysed alongside five laboratory standards spanning the range of expected isotope values and international reference standards to correct for daily variation. Measured delta values were subsequently converted to deuterium concentration (ppm). O^{18} results were also obtained.

2.4. Statistical analysis

Analysis included only houseflies for which death was classified as age-related. Analysis excluded any non-eclosion ($n = 34$) or those that died stuck in cotton wool at the top of vial ($n = 68$). Statistical analyses were conducted in RStudio 2026.01.1 + 403 with ggplot 2 (version 3.5.2) and dplyr (version 1.1.4). Survival was analysed using a discrete-time survival modelling framework. Individual-level survival records were expanded into a fly-day dataset, with one row per individual for each day that it remained at risk. The response variable indicated whether death occurred on a given day (death = 1) or whether the individual survived that day (death = 0). We fitted a generalised linear model with a complementary log-log link, which is appropriate for modelling discrete-time hazards and estimates the daily probability of death conditional on survival to the start of that day. Age-dependent changes in baseline mortality were modelled flexibly using a natural spline for day with five degrees of freedom. DDW concentration was included as standardised log-transformed ppm, with sex, standardised body weight, and the interaction between DDW concentration and sex included as predictors. Thus, the model tested whether DDW concentration affected daily mortality risk, whether this effect differed between males and females, and whether body weight at time of death explained additional variation in survival after accounting for age, sex, and DDW concentration. Kaplan–Meier survival curves were also estimated to visualise empirical survival patterns and were overlaid with model-predicted survival curves from the discrete-time model. Model predictions were generated at mean body weight for each concentration-by-sex combination and converted from daily death probabilities into cumulative survival probabilities. Analysis were generated using the "survival" package version 3.8.3 and the "survminer" package version 0.5.1.

3. Results

3.1. Isotope enrichments in the water, dietary water and the flies' bodies

We measured the deuterium and ^{18}oxygen enrichments in the prepared water, water extracted from the diet and that in body water of the flies compared to the nominal enrichments based on the suppliers claims about the enrichments (Table 1). Aberdeen tap water had an enrichment of 147 ± 0.04 ppm deuterium and 1982 ± 0.2 ppm ^{18}oxygen consistent with a latitude of 57 °N. The actual measured enrichments of the depleted waters and diet strongly correlated with the target enrichments. The ^{18}oxygen level was stable at 1981–1982 ppm in prepared water and 1999 in the diet. The body water enrichment in the flies when feeding on the natural abundance water diet was higher for both isotopes (161 ± 0.1 ppm for deuterium and 2000 ± 0.1 ppm for ^{18}oxygen). The other enrichments followed the same pattern as in the diet, but the effects of depletion and enrichment in the supplied water were attenuated. Hence the lowest level in the diet (48 ppm) led to an enrichment in the bodies of the flies of 67 ppm, while at the highest level in the diet (949 ppm) the enrichment in the body water of the flies was only 808 ppm (Table 1). Deuterium levels in water and dietary water strongly correlated with the target levels and those found in houseflies (Fig. S1).

3.2. Lifespan effects of dietary deuterium ranging from 48 to 950 ppm

In total 491 pupae eclosed (6.5% non-eclosion) and 68 died of non-age related causes (see methods). Sex and weight data was not available for 3 animals. These were not included in the analysis. Survival analysis was therefore carried out using data for 420 houseflies.

Median survival was not significantly different over the range of deuterium levels (log-rank test $\chi^2 = 3.2$, df = 6, $p = 0.8$; Fig. 1a, Table 2). Survival did however differ between the sexes and females exhibited significantly longer lifespans than males ($\chi^2 = 47.9$, df = 1, $p = 4 \times 10^{-12}$; Fig. 1b). Median lifespan in the control (diet = 150 ppm) was 56 days (95% CI 53–61 days). The median lifespan in flies fed diets at the lowest deuterium level, 48 ppm, was 53 (50–61) days and highest level, 950 ppm, was 57 (52–61) days. Females were significantly heavier than males (F = 327.21, $p = 2 \times 10^{-1}6$; Fig. 1c). However, body weight at time of death, and presumably feeding, was not affected over the range of deuterium concentrations tested (F = 0.081, $p = 0.78$; Fig. 1c). No relationship between body weight at time of death or survival was noted (Females: $R = -0.12$, $p = 0.073$, males: $R = -0.069$, $p = 0.34$; Fig. 1d).

The discrete-time survival model showed no evidence that deuterium abundance affected survival after accounting for age-dependent mortality, sex, and body weight (Fig. 2; Table 3). Daily mortality risk changed nonlinearly with age, as captured by the natural spline fitted to day. Spline coefficients represent basis functions rather than discrete biological time periods and thus, were interpreted jointly as evidence that the baseline hazard varied through time. There was no evidence for an effect of deuterium abundance on daily mortality risk among females, which form the reference sex in the model (estimate = −0.005, hazard ratio = 0.995, $p = 0.936$). Males had higher daily mortality risk than females after controlling for the other predictors, with an estimated 2.10-fold increase in the discrete-time hazard scale (estimate = 0.744, hazard ratio = 2.10, $p < 0.001$). Body weight at death was not associated with daily mortality risk (estimate = 0.018, hazard ratio = 1.02, $p = 0.804$). There was also no evidence that the effect of deuterium abundance differed between females and males (DLW concentration x sex interaction: estimate = 0.005, hazard ratio = 1.005, $p = 0.960$). Overall, survival differences were explained primarily by age-dependent mortality and sex, with no detectable contribution of body weight, or sex-specific responses to deuterium.

4. Discussion

Levels of deuterium in the body water of the flies were consistent with those supplied in the diet. When feeding on the diet made up with Aberdeen tap water, the body water enrichment in the flies was higher for both isotopes, 161 ppm for deuterium compared to 147 ppm in the dietary water and 2000 ppm for ^{18}oxygen compared to 1981 ppm in the dietary water. This enrichment effect is because the body water of the flies derives from a combination of the dietary supplied water, water

Table 1
Isotopic levels of water used for diet preparation, water extracted from the diet and in the body water of houseflies feeding on those diets. Data shown as mean of 5 measures ± SD.

Target ppm	Water ^{2}H	Water ^{18}O	Diet ^{2}H	Diet ^{18}O	Houseflies ^{2}H	Houseflies ^{18}O
25	31 ± 0.1	1800 ± 0.1	48 ± 0.03	1741 ± 0.1	67 ± 0.2	1788 ± 0.1
50	54 ± 0.1	1740 ± 0.2	66 ± 0.1	1782 ± 0.1	90 ± 0.1	1839 ± 0.1
100	101 ± 0.05	1863 ± 0.3	109 ± 0.1	1891 ± 0.1	118 ± 0.1	1909 ± 0.1
150	147 ± 0.04	1982 ± 0.2	150 ± 0.1	1983 ± 0.1	161 ± 0.1	2000 ± 0.1
250	248 ± 0.2	1981 ± 0.3	246 ± 0.2	1997 ± 0.1	230 ± 0.2	1996 ± 0.05
500	506 ± 0.4	1982 ± 0.3	478 ± 0.3	1999 ± 0.1	432 ± 0.7	2002 ± 0.5
1000	1021 ± 0.9	1982 ± 0.3	950 ± 0.5	1999 ± 0.2	808 ± 0.7	2001 ± 0.6

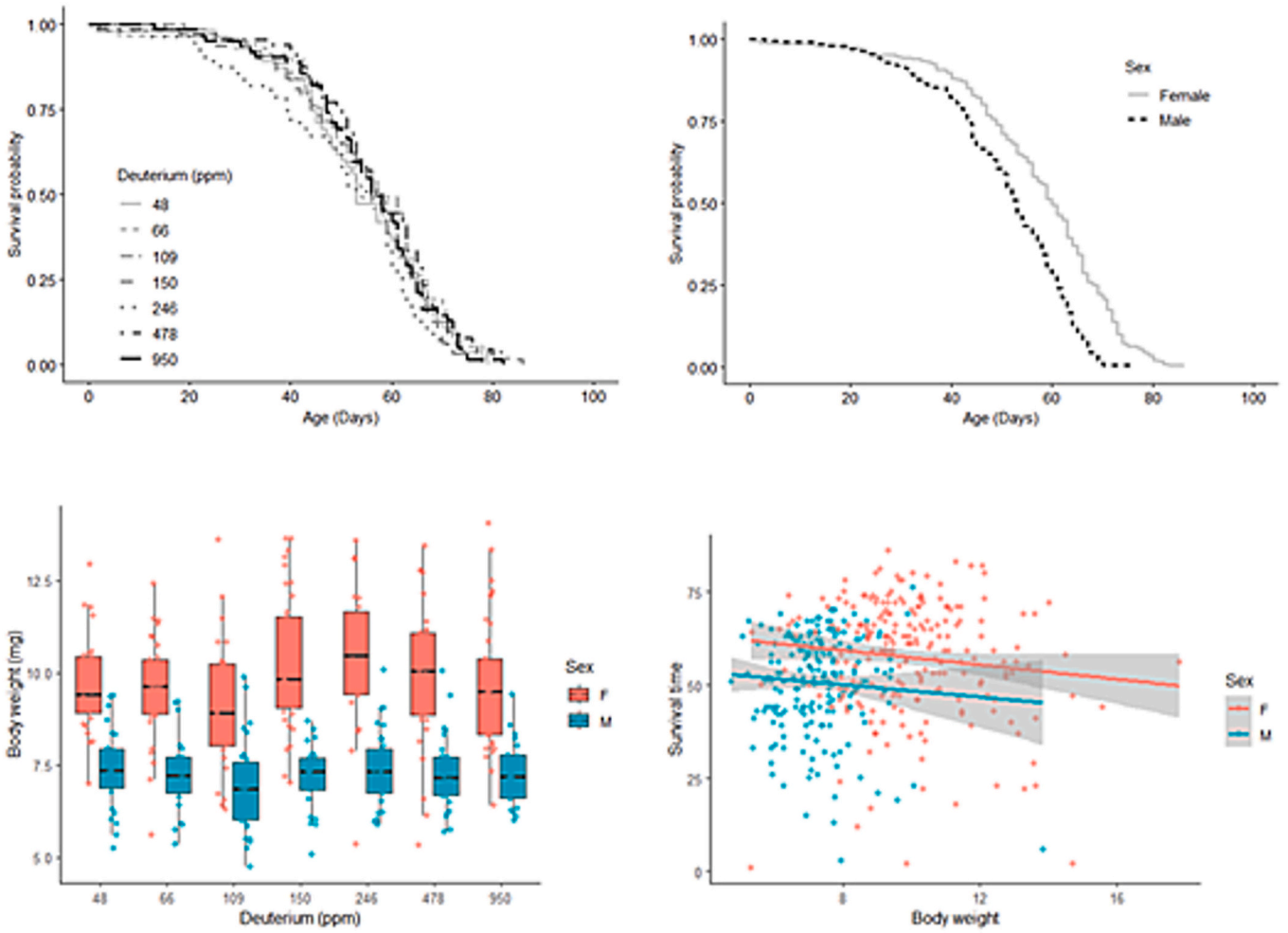


Fig. 1. Survival and body weight at death of male and female houseflies across deuterium abundance ranging from 48 to 950 ppm in water extracted from diets. a) No significant effect of deuterium concentration on lifespan was detected ($p = 0.8$), b) females exhibited significantly longer lifespans than males ($p = 4 \times 10^{-12}$), c) females were heavier than males, and d) body weight at death was not related to lifespan. Total $n = 420$ (225 females, 195 males).

uptake from the atmosphere and metabolic water generated from metabolising the food. Moreover, there is a fractionation effect when water is lost by evaporation leading to enrichment of the isotopes left behind (Kreuzer-Martin et al., 2006). These processes also attenuated the effects of the other dietary water manipulations. Hence the body water contents of the flies regressed towards the natural abundance enrichment. When feeding on depleted waters the enrichments in the flies were higher, and when feeding on enriched waters the enrichments were lower than in the diet. Nevertheless, the enrichment of the housefly body water when feeding on the lowest enrichment diet (48 ppm diet; 67 ppm body water) was less than 1/10th of the enrichment of the water in the bodies of flies feeding on the highest enrichment diet (950 ppm diet; 808 ppm body water). So, while the manipulation was attenuated there were still large differences between groups which cover the range where previous studies have suggested there are large health impacts. One way that deuterium enrichment could affect lifespan is if deuterium in the diet affected food intake. This would be unlikely to be a direct response to the deuterium itself as we are not aware of any mechanism by which flies (or indeed any animal) can sense the levels of deuterium in their diet. However, there could potentially be an indirect impact via kinetic isotope effects downstream of ingestion – e.g. on uptake rates etc., or regulatory processes for food intake. Measuring food intake of individual flies is challenging (Wong et al., 2008; Deshpande et al., 2014). A consistent marker of calorie restriction, however, is altered

Table 2
Dietary deuterium supplementation over the range of 48–950 ppm had no effect on lifespan of houseflies (*Musca domestica*). Diets prepared with 3 deuterium depleted and 3 deuterium enriched levels were compared to a tap water control group (147 ppm). Seventy five pupae were allocated per treatment group. Body water of houseflies (ppm) shown. N = individuals for which survival analysis was completed. Data shown as days. LCL and UCL = lower and upper confidence limit.

Houseflies	N	Mean	SE	Median	0.95 LCL	0.95 UCL
67	55	53.5	2.0	53	50	61
90	57	54.7	2.0	54	51	61
118	63	55.4	1.7	58	53	63
161	63	53.9	1.8	56	53	61
230	55	50.3	2.3	56	50	59
432	65	56.6	1.8	57	54	63
808	62	55.2	1.7	57	52	61

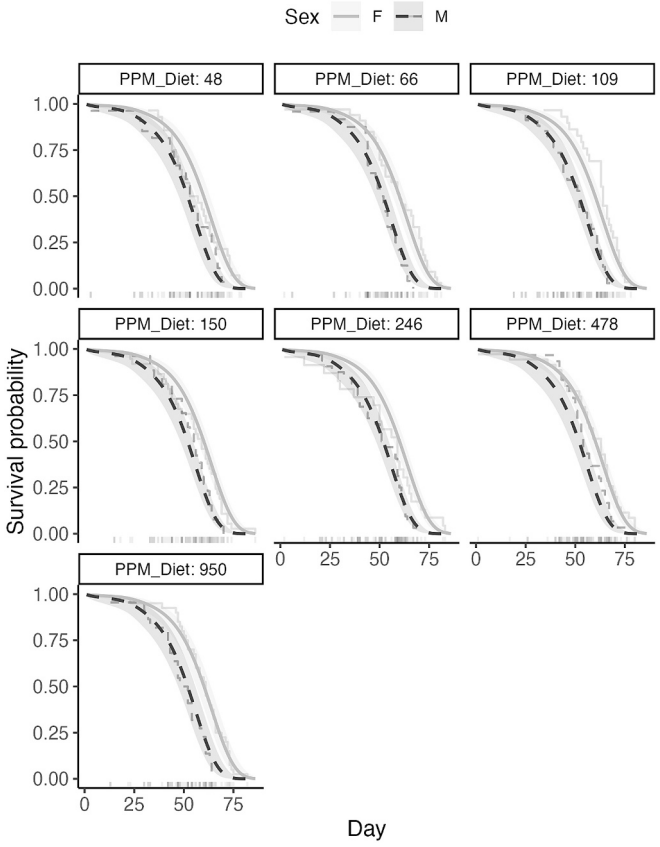


Fig. 2. Survival probability of flies across deuterium abundance ranging from 48 to 950 ppm in water extracted from the diet. Kaplan–Meier survival curves are shown for females and males at each measured concentration of deuterium in water extracted from the diet, expressed in ppm. Solid light-grey lines indicate females and dashed dark-grey lines indicate males. Smooth curves show survival predicted from the final discrete-time survival model with a complementary log-log link, including age, sex, body weight at death, deuterium abundance in diet-extracted water, and the interaction between deuterium abundance and sex. Shaded ribbons show 95% confidence intervals around model predictions. Grey step curves show the observed Kaplan–Meier estimates, and rugs along the x-axis show observed death times. Facets correspond to the measured deuterium abundance in water extracted from the diet. Predictions are shown at mean death body weight.

body weight. For example, in mice the average weight loss (and hence final weight) is directly related to the level of restriction (Mitchell et al., 2015). Here, no differences in body weight at death of male or female flies in relation to the level of deuterium in the dietary water was observed, suggesting the dietary deuterium levels did not precipitate any indirect effects via modulation of food intake. Mortality was also unrelated to body weight at death, further supporting the interpretation that there were no indirect impacts via food intake.

Table 3
Deuterium (^{2}H) abundance in dietary water was not associated with survival in houseflies across the tested range of 48–950 ppm. The table reports a discrete-time survival model with a complementary log-log link, in which individual survival records were expanded into fly-day observations and death was modelled as a conditional daily event. The final model included a five-degree-of-freedom natural spline for age, standardised log-transformed ^{2}H ppm, sex, body weight, and the interaction between ^{2}H and sex.

Term	Estimate	Std.error	Statistic	p.value
(Intercept)	−6.41	0.61	−10.43	0.00
ns(day, df = 5)1	0.59	0.69	0.86	0.39
ns(day, df = 5)2	1.19	0.67	1.78	0.07
ns(day, df = 5)3	3.85	0.43	9.01	0.00
ns(day, df = 5)4	3.36	1.55	2.16	0.03
ns(day, df = 5)5	6.58	0.45	14.57	0.00
log_ dietary ^{2}H_z	−0.01	0.07	−0.08	0.94
SexM	0.74	0.13	5.54	0.00
Weight_z	0.02	0.07	0.25	0.80
log_ dietary ^{2}H_z:SexM	0.00	0.10	0.05	0.96

Despite many previous studies suggesting positive impacts of DDW (in cell cultures and on mice) (Cong et al., 2010; Somlyai et al., 2022) we did not find any evidence for a lifespan effect with deuterium isotope enrichments over the range 67 to 808 ppm in the body water, consequent of imbibing water with levels between 31 and 1021 ppm. The isotopic resonance model suggests there are certain "resonance" abundances of stable isotopes, at which a compositional symmetry emerges. This is suggested to reduce the complexity of the system built of these elements, affecting the kinetics of reactions within the system. While we recognise the specific combination of isotopes C, H, N, and O proposed by isotopic resonance was not tested, the deuterium levels tested here include those suggested in an isotopic resonance effect (Rodin et al., 2018). If this does happen in flies, we found no evidence that it has important impacts as far as the housefly lifespan is concerned.

Previous reports indicated an increase in lifespan of *Drosophila* at levels of deuterium above background (Hammel et al., 2013). Several differences may explain these contrasting results. Our study differed from that of Hammel in the period of deuterium exposure. While we opted to study the adult phase, Hammel et al. (Hammel et al., 2013) exposed eggs directly to deuterated food and once eclosed the adult flies (~11 days) were then removed to non-deuterated diets. Critically the levels of deuterium reported to increase lifespan in the *Drosophila* were also considerably higher than we studied, 7.5 and 15% versus maximally 0.1% in our study.

4.1. Limitations

We only exposed flies to the variable levels of deuterated water in their diets after the larval stage was complete. Previous work showed higher deuterium levels than we used, added to diet at initial development stages extended lifespan in *Drosophila* (Hammel et al., *2013*). It is possible therefore that there are impacts on earlier developmental stages that were missed by our protocol. Moreover, a potential argument for why we found no relationship between the deuterium enrichment and the lifespan is that the main mechanism by which DDW is suggested to impact health and longevity is by interfering with cell division (Zhang et al., 2020; Yavari and Kooshesh, 2019). Because adult flies are mostly post-mitotic then one might not expect strong benefits of deuterium depletion on lifespan. However, contrasting this viewpoint, while adult flies don't grow after metamorphosis, they still have proliferative cells, and *Drosophila* are a well-established model for cancer research (Villegas, 2019; Yamamura et al., 2021). Moreover, the anti-cancer effect is only 1 of 4 different mechanisms by which DDW is

presumed to influence health and lifespan. In particular there are suggestions that it reduces oxidative stress and that might be expected to impact longevity independent of any cancer susceptibility effects (Zhang et al., 2020; Bayrak et al., 2022).

5. Conclusion

Using the housefly as a model, we found no evidence that dietary deuterium supplementation over the range of 31–1021 ppm influenced growth or survival. These findings contrast with previous reports of beneficial effects observed across a wide range of deuterium abundance (30–300,000 ppm). Our results suggest that responsiveness to interventions targeting ageing may not be conserved across species and any longevity-promoting effects of dietary deuterium are unlikely to be universal.

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CRediT authorship contribution statement

Audrey Riddle: Investigation. **Charlie Hinson:** Investigation. **Peter Thomson:** Validation, Methodology. **Catherine Hambly:** Validation, Methodology. **Juliano Morimoto:** Writing – review & editing, Formal analysis. **John R. Speakman:** Writing – review & editing, Conceptualization. **Sharon E. Mitchell:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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